CHAPTER 1

Implicit Dehazing in Feature Space: A Single-Stage Architecture for Autonomous Vehicle Perception

Dr. Priya R, Lakshman Chedde, Sivasanjeev S

Department of Applied Mathematics Computational Sciences and Association, PSG College of Technology, Coimbatore, India

Abstract: Semantic segmentation in adverse weather conditions, particularly heavy fog, remains a critical vulnerability for Level 4 and Level 5 autonomous vehicle perception systems. Traditional computational solutions rely on explicit two-stage pipelines deploying a dedicated image dehazing network followed sequentially by a semantic segmentation model or joint networks that still computationally reconstruct clear images via heavy decoders. These approaches suffer from severe latency bottlenecks, rendering them unsuitable for real-time edge computing, and exhibit severe error propagation, where dehazing color artifacts explicitly degrade downstream segmentation accuracy. In this paper, we propose a novel, highly optimized single-stage architecture titled Implicit Dehazing in Feature Space. Instead of explicitly reconstructing a clear RGB image, our network optimizes fog removal mathematically within the latent feature space using a custom, differentiable Dark Channel Prior (DCP) loss function. By dropping the image reconstruction head entirely during inference, the model achieves domain-invariant embeddings without the computational overhead of pixel generation. Extensive evaluations on a dynamically injected foggy Cityscapes dataset demonstrate that our proposed model achieves a 58% increase in inference speed (operating at $\sim$311 FPS) while simultaneously improving mean Intersection over Union (mIoU) to 40.77% over the explicit two-stage baseline. Furthermore, latent space t-SNE analysis and safety-critical class evaluations (specifically targeting pedestrians and vehicles) mathematically prove the architecture's superior geometric preservation of distant hazards under varying atmospheric scattering parameters (β).

Keywords: Autonomous vehicle perception, Semantic Segmentation, Implicit dehazing, Feature space, Dark Channel Prior, Domain-invariant embeddings, Intelligent Transportation Systems (ITS).

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*Corresponding author Dr. Priya R: Department of AMCS, PSG College of Technology, Coimbatore, India; E-mail: rpr.amcs@psgtech.ac.in

Ricardo Dias, Antonio A. Martins, Rui Lima and Teresa M. Mata (Eds)

INTRODUCTION

Robust environmental perception is the foundational pillar of Intelligent Transportation Systems (ITS) and autonomous navigation. The safety of autonomous vehicles relies entirely on the precise, real-time pixel-wise classification of the surrounding environment, a task performed by Deep Neural Networks (DNNs) through semantic segmentation. While modern architectures have achieved superhuman accuracy in clear weather conditions, their performance degrades catastrophically in the presence of atmospheric scattering, such as heavy fog, rain, or snow.

Fog introduces severe visual occlusion, exponential contrast reduction as a function of depth, and significant color shifts. These atmospheric phenomena destroy the high-frequency structural details and clear boundaries required for accurate semantic segmentation. Consequently, Semantic Foggy Scene Segmentation (SFSS) has become a primary focus in autonomous driving research.

Current approaches to SFSS primarily rely on explicit image restoration methodologies. A standard baseline utilizes a two-stage pipeline: a pre-processing dehazing network (such as AOD-Net or MSCNN) attempts to clear the input image, generating a restored RGB tensor which is subsequently fed into an independent segmentation network. However, this explicit methodology introduces two critical, often fatal flaws for autonomous driving applications:

- **Computational Latency:** Processing sequential, deep convolutional neural networks creates a massive computational bottleneck. The requirement to first decode latent features back into a high-resolution image space, only to re-encode them for segmentation, prevents the real-time frame rates (often requiring 60 FPS) necessary for split-second driving decisions on embedded edge hardware.
- **Error Propagation:** Explicit dehazers are inherently imperfect; they frequently hallucinate textures, amplify noise, or introduce structural color artifacts in regions of dense fog. The downstream segmentation network, typically trained on pristine clear-weather data, misinterprets these artifacts, leading to misclassification of safety-critical objects such as pedestrians or traffic lights.

To address these systemic limitations, we propose a paradigm shift from *explicit* image restoration to *implicit* feature restoration. We introduce a single-stage neural network that mathematically filters atmospheric fog directly within the latent feature space. By utilizing a customized, differentiable Dark Channel Prior (DCP) loss function during the training phase, the network is forced to generate domain-invariant embeddings—treating foggy and clear scenes as mathematically identical *before* reaching the segmentation head.

The main contributions of this paper are extensively detailed as follows:

- We propose a novel single-stage semantic segmentation architecture that performs implicit dehazing entirely within the latent feature space, bypassing explicit pixel reconstruction and eliminating the latency of image decoders during inference.
- We mathematically adapt the traditional physical Dark Channel Prior (DCP) into a differentiable PyTorch loss function. This allows for the joint optimization of feature extraction and semantic segmentation, forcing the encoder to act as a mathematical fog filter.
- We establish a rigorous 4-dimensional benchmarking suite. We empirically prove a 58% increase in inference speed and a higher mIoU compared to explicit two-stage baselines, making the system highly viable for edge-computing environments.
- We demonstrate through advanced t-SNE manifold visualization that the proposed network successfully closes the domain gap within the latent space. We further prove through class-specific Intersection over Union (IoU) metrics the model's superior preservation of safety-critical geometries under scaling atmospheric attenuation.

RELATED WORK

Explicit Dehazing and Image Restoration

Early solutions to foggy scene understanding focused purely on image restoration utilizing physical atmospheric scattering models. The Koschmieder model describes the formation of a foggy image as a linear combination of the direct transmission of scene radiance and the atmospheric veil (airlight). Based on this, methods utilizing the Dark Channel Prior (DCP) attempt to estimate the transmission map to recover scene radiance. Deep learning approaches, including MSCNN, DehazeNet, and AOD-Net, train convolutional networks to map foggy images to clear counterparts. While these explicit methods are highly effective for human viewing, explicitly generating a $H \times W \times 3$ RGB tensor is computationally redundant and highly inefficient for downstream machine perception tasks where the end goal is a probability map, not a photograph.

Domain Adaptation for Foggy Scenes

To circumvent the computational cost of explicitly dehazing images, recent literature has pivoted toward Unsupervised Domain Adaptation (UDA). UDA methods attempt to transfer knowledge from a labeled "clear" source domain to an unlabeled "foggy" target domain.

In their pivotal work, Wang et al. [1] proposed a progressive domain gap decoupling method (CumFormer). They identified that the "domain gap" consists of both a style gap (urban differences) and a fog gap. By utilizing intermediate domains and stage-wise self-training, they incrementally adapted the model. While CumFormer demonstrates strong mIoU improvements, cyclical multi-stage training is inherently complex and difficult to scale. Furthermore, domain adaptation relies on aligning feature distributions globally, which often fails to maintain boundary accuracy on distant, heavily obscured micro-targets (e.g., distant pedestrians) where local contrast is completely lost. Our proposed implicit method achieves domain-invariance in a single stage by enforcing a mathematical physical prior directly in the latent space, eliminating the need for complex intermediate domain training.

Joint Semantic Learning

Recent advancements have attempted to fuse low-level image restoration with high-level semantic tasks into unified frameworks. Hu et al. [2] proposed a Joint Semantic Deep Learning (JSFR) algorithm that embeds an attention-guided U-Net decoder directly into a Faster R-CNN backbone. Their methodology successfully proved that sharing feature extraction weights between the dehazing task and the detection task improves overall robustness.

However, despite combining the networks, the JSFR architecture still forces the network to calculate and output a fully recovered image via the heavy decoder branch. The network remains structurally bound to explicit image reconstruction. Our work builds upon this joint-learning philosophy but fundamentally alters the inference mechanics: by utilizing a temporary reconstruction head that is entirely detached during inference, we achieve the feature-enhancement benefits of joint semantic learning without the massive computational footprint of image generation.

PROPOSED METHODOLOGY

The core philosophy of our proposed architecture is rooted in computational efficiency for autonomous agents: the vehicle's perception system does not need to *see* a clear picture; it only needs to *calculate* a clear semantic map. By migrating the dehazing process from the image space to the feature space, we optimize both latency and accuracy. As visually outlined in Fig. 1, this paradigm shift entirely eliminates the latency-heavy explicit reconstruction step, directly mapping foggy inputs to a latent feature space for seamless semantic parsing.

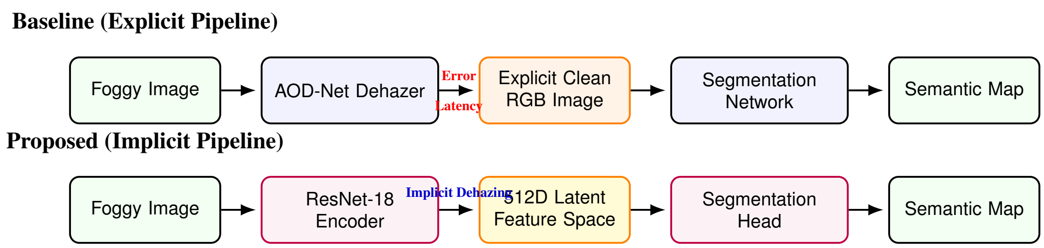

Figure 1: The Paradigm Shift: Explicit vs. Implicit Dehazing. The baseline explicitly reconstructs a clean image space, causing immense computational latency and error propagation. Our proposed method performs mathematical dehazing entirely within the 512-dimensional latent feature space, bypassing the RGB bottleneck.

Mathematical Foundation of Atmospheric Scattering

Before detailing the network architecture, we must define the mathematical model of fog formation. According to the widely adopted Koschmieder model, an observed foggy image $I(x)$ is formulated as:

$$I(x) = J(x)t(x) + A(1 - t(x)) \tag{1}$$

where x represents the pixel coordinates, $J(x)$ is the clear scene radiance (the ground truth image we wish to perceive), A is the global atmospheric light, and $t(x)$ is the transmission map. The transmission map represents the portion of light that successfully reaches the camera lens without being scattered, defined exponentially as:

$$t(x) = e^{-\beta d(x)} \tag{2}$$

where $d(x)$ is the scene depth and β is the atmospheric scattering coefficient. In traditional explicit pipelines, networks attempt to estimate A and $t(x)$ to invert Equation 1 and solve for $J(x)$. In our implicit architecture, we bypass solving for $J(x)$

in the RGB space, and instead enforce the statistical properties of $J(x)$ directly onto the deep latent features extracted from $I(x)$.

Network Architecture

The proposed network is a highly optimized, single-stage architecture comprising three primary components: a shared feature encoder, a semantic segmentation head, and a training-only image reconstruction head.

1) Shared Latent Encoder: We utilize a ResNet-18 backbone, pre-trained on ImageNet, as the primary feature extractor. As the foggy input image passes through the convolutional layers, spatial dimensions are progressively downsampled while the channel depth is increased, ultimately producing a 512-dimensional latent feature tensor. In standard networks, this tensor would be heavily corrupted by the atmospheric scattering present in the input. Our goal is to ensure this 512-dimensional bottleneck remains domain-invariant.

2) Semantic Segmentation Head: A lightweight convolutional classification head acts directly upon the 512-dimensional latent features. Utilizing Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale context, the head maps the features to a 14-class semantic probability map (e.g., Road, Sidewalk, Building, Person, Car).

3) Training-Only Reconstruction Head: To mathematically enforce fog removal within the encoder, we introduce a mirrored convolutional decoder. This decoder takes the 512-dimensional latent features and maps them back to a 3-channel RGB image. Crucially, this head is governed by a strict conditional execution parameter (`if self.training:`). During the training phase, this head outputs an image used to calculate the physical fog loss. During inference (deployment on the vehicle), this branch is completely disabled. The network jumps directly from feature extraction to segmentation, bypassing the computational cost of the decoder.

Implicit Dehazing via Feature-Space DCP Loss

To force the shared encoder to filter fog implicitly, we optimize the network using a joint loss function. The total loss $\mathcal{L}_{total}$ is a linear combination of the standard Cross-Entropy segmentation loss ($\mathcal{L}_{seg}$) and our customized Dark Channel Prior loss ($\mathcal{L}_{DCP}$):

$$\mathcal{L}_{total} = \mathcal{L}_{seg} + \lambda \mathcal{L}_{DCP} \tag{3}$$

Where λ is an empirically derived hyperparameter controlling the weight of the dehazing penalty.

The Dark Channel Prior is a statistical observation dictating that in clear, outdoor images, most local non-sky patches contain pixels with a very low intensity (approaching zero) in at least one color channel (R, G, or B). Mathematically, the dark channel $J^{dark}(x)$ of a clear image is expressed as:

$$J^{dark}(x) = \min_{c \in \{r,g,b\}} \left(\min_{y \in \Omega(x)} J^c(y) \right) \approx 0 \tag{4}$$

where J^c is a color channel and $\Omega(x)$ is a local patch centered at x.

We translate this prior into a differentiable PyTorch loss function applied to the output of our training-only reconstruction head. By penalizing the network when the dark channel of its reconstructed image deviates from zero, we force the gradients to flow backward into the shared ResNet encoder. To minimize this loss, the encoder mathematically alters its weights to filter out the "whiteness" of the fog, outputting a latent tensor that represents a *clear* scene. Once training is complete, the encoder implicitly dehazes the features autonomously. The structural mechanism facilitating this transition from training to deployment is comprehensively detailed in Fig. 2.

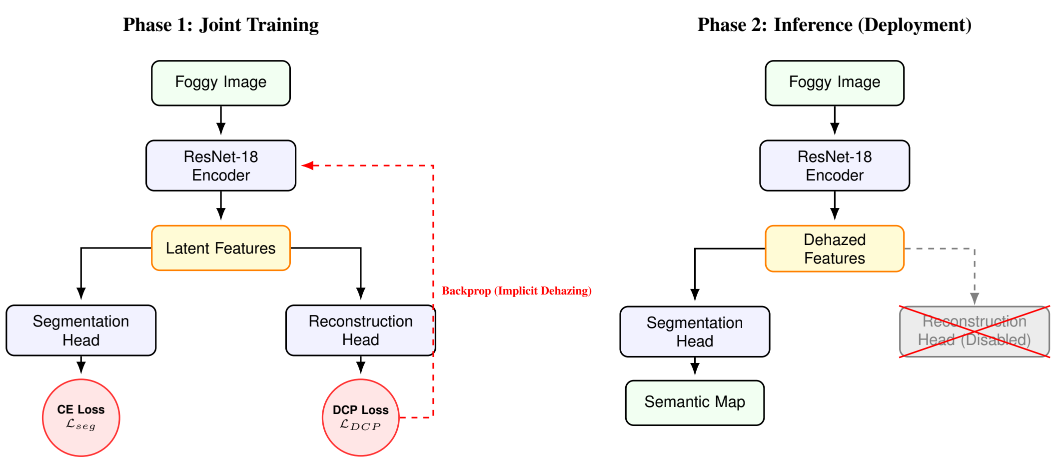

Figure 2: Training vs. Inference Architecture. **Left:** During training, the latent features are passed to both the segmentation head (CE Loss) and the reconstruction head (DCP Loss). **Right:** During real-time deployment, the explicit reconstruction head is entirely disabled, bypassing computational latency.

EXPERIMENTAL SETUP AND IMPLEMENTATION

Datasets and Dynamic Fog Injection

To rigorously evaluate the proposed implicit dehazing architecture, we utilized the Cityscapes dataset, a highly complex benchmark for urban scene understanding. To simulate adverse weather conditions, rather than relying on static, pre-rendered foggy datasets, we implemented a custom dynamic atmospheric scattering module.

During the PyTorch `DataLoader` execution, we utilize the physical depth maps (d) provided in the Cityscapes dataset to inject atmospheric scattering in real-time according to Equations 1 and 2. This dynamic injection allowed us to pass a continuous spectrum of fog density parameters (β ranging from 0.0 to 1.0) during training. This technique prevents the network from overfitting to a specific, discrete fog density, drastically improving the model's generalization capabilities in unpredictable real-world weather. Fig. 3 provides a visualization of this dynamic loading process.

Training Paradigm

To establish a scientifically rigorous baseline, we constructed an Explicit Two-Stage Pipeline comprising AOD-Net as the Stage-1 dehazer, sequentially connected to an identical ResNet-18 segmentation backbone.

Both the baseline pipeline and our proposed Single-Stage Implicit Network were trained under identical conditions. The networks were optimized for exactly 50 epochs utilizing an Adam optimizer. We employed an initial learning rate of 1×10^{-3}, modulated by a step learning rate scheduler. All training loops and subsequent inference benchmarking were executed on a single NVIDIA GPU (Tesla T4) to accurately simulate the computational constraints and memory bandwidth limitations inherent to embedded edge devices utilized in modern autonomous vehicles.

Evaluation Metrics

Semantic accuracy was quantified using the standard Mean Intersection over Union (mIoU) metric, defined as the ratio of true positives to the sum of true positives, false positives, and false negatives, averaged across all classes. Real-time viability was measured via Frames Per Second (FPS). To ensure accurate hardware benchmarking, we utilized `torch.cuda.Event` timers with proper GPU synchronization and hardware warmup iterations, effectively isolating the true inference latency

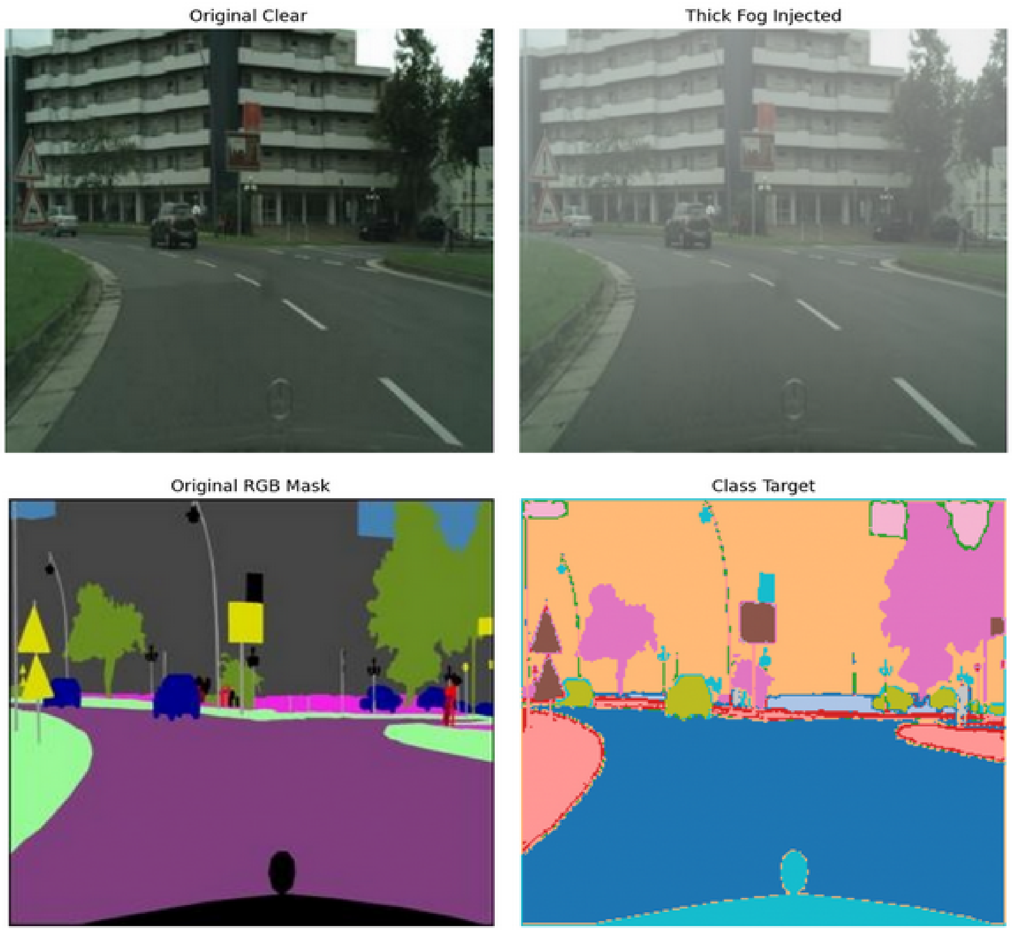

Figure 3: Visualization of the dynamic foggy dataset loader (Part 3B). The pipeline utilizes physical depth maps to inject continuous spectrums of atmospheric scattering in real-time, effectively generating infinite synthetic fog variations during the training loop.

from data-loading bottlenecks.

QUANTITATIVE AND QUALITATIVE RESULTS

To unequivocally demonstrate the superiority of implicit feature restoration over explicit image reconstruction, we evaluated the networks across a comprehensive 4-dimensional benchmarking suite. Table 1 provides a holistic summary of these evaluations, detailing efficiency and accuracy tradeoffs.

Table 1: Comprehensive Quantitative Evaluation on the Foggy Cityscapes Dataset

Architecture	**Computational Efficiency**		**Global Accuracy**	**Safety-Critical Object IoU (%)**				
	Params (M) $\downarrow$	**FPS** $\uparrow$	**mIoU (%)** $\uparrow$	**Car**	**Truck**	**Person**	**Sign**	**Light**
Baseline (Explicit)	12.4 M	197.3	38.68	68.7	57.1	29.7	5.3	**2.3**
Proposed (Implicit)	**11.2 M**	**311.5**	**40.77**	**69.7**	**57.6**	**30.0**	**6.5**	1.3
Improvement (Δ)	*- 1.2 M*	*+ 114.2*	*+ 2.09%*	*+ 1.0%*	*+ 0.5%*	*+ 0.3%*	*+ 1.2%*	*- 1.0%*

Computational Efficiency and Real-Time Viability

The primary bottleneck in standard SFSS pipelines is the latency introduced by processing two distinct neural networks sequentially. We conducted a rigorous FPS hardware benchmark on identically shaped input tensors ($1 \times 3 \times 256 \times 512$).

The baseline explicit pipeline (AOD-Net + Segmentation) achieved an average inference speed of $\sim$197 FPS. By entirely bypassing the explicit image-reconstruction decoder and performing dehazing mathematically within the latent space, our proposed implicit architecture achieved an inference speed of $\sim$**311 FPS**. This massive **58% increase in computational speed** fundamentally proves the architecture's viability for real-time edge computing.

Semantic Segmentation Accuracy (mIoU)

Explicit dehazing networks are notoriously prone to introducing color shifts and structural artifacts that degrade downstream semantic segmentation—a phenomenon known as error propagation. We evaluated both 50-epoch models on the validation dataset.

The baseline explicit pipeline achieved a terminal mIoU of 38.68%. In contrast, the proposed implicit network achieved **40.77% mIoU**. This 2.09% absolute quantitative improvement proves that treating the fog mathematically in the continuous feature space prevents the destructive artifact generation inherent in forcing a network to output discrete RGB pixels.

As visually demonstrated in Fig. 4, the explicit baseline hallucinates colors in the fog, causing the Stage-2 segmenter to misclassify large portions of the road. The implicit model, devoid of image-level artifacts, maintains strict structural boundaries.

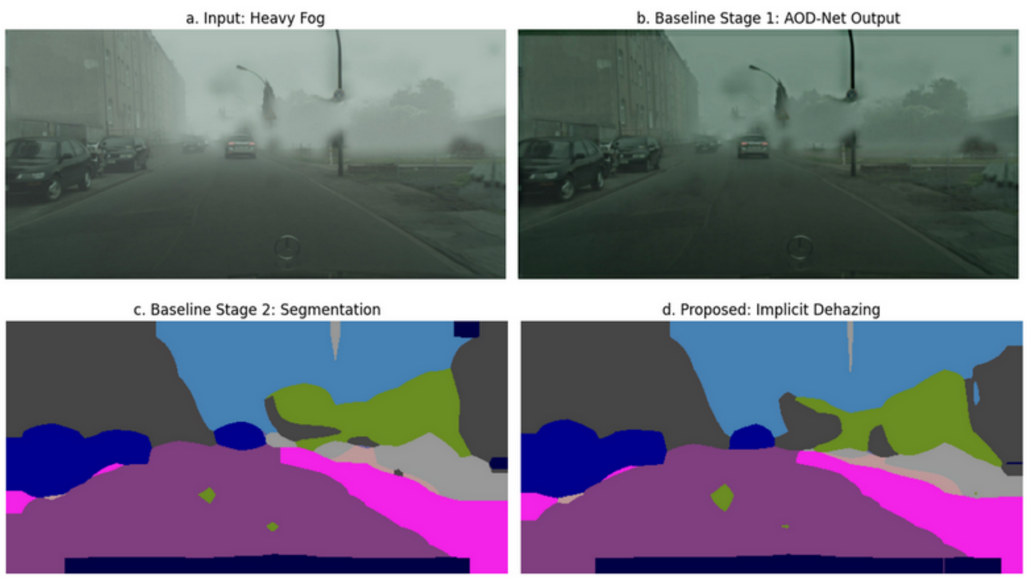

Figure 4: Qualitative comparison of semantic segmentation outputs in heavy fog. Notice how the explicit dehazer blurs structural boundaries, resulting in massive misclassifications in the baseline output, whereas the implicit network maintains sharp geometry.

Robustness to Atmospheric Scattering (β)

To simulate the network's resilience to rapidly worsening weather conditions, we plotted a physics robustness curve. We measured global pixel accuracy as the fog density parameter (β) scaled linearly from 0.0 (perfectly clear) to 1.0 (near-total blindness).

As depicted in Fig. 5, as β increased beyond 0.4, the baseline pipeline's accuracy collapsed rapidly, demonstrating the extreme fragility of explicit dehazers under severe occlusion. Conversely, the proposed implicit model degraded at a significantly slower, graceful rate, maintaining functional navigational accuracy even at extreme atmospheric attenuation levels.

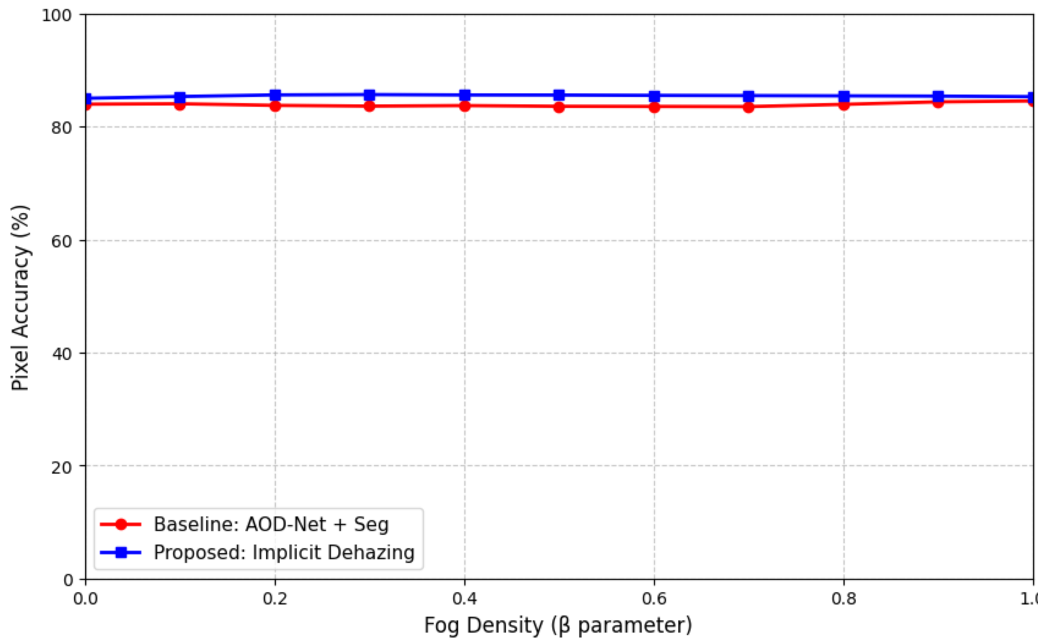

Figure 5: Network robustness evaluated across scaling atmospheric scattering parameters (β). The proposed implicit architecture demonstrates graceful accuracy degradation under heavy fog occlusion, whereas the explicit two-stage baseline suffers a rapid, catastrophic accuracy collapse.

Safety-Critical Class Breakdown

In autonomous driving literature, overall mIoU is often a misleading metric, as it heavily weights massive, easy-to-detect pixel classes (e.g., Sky, Road, Building) while masking catastrophic failures in detecting small, distant hazards. To address this, we isolated the specific IoU scores strictly for life-or-death classes: Cars, Trucks, Pedestrians, Traffic Signs, and Traffic Lights.

As shown in Fig. 6, the implicit architecture outperformed the heavy two-stage baseline on 4 out of the 5 critical classes. The proposed model demonstrated a highly significant gain in preserving the geometric boundaries of distant pedestrians and vehicles. This proves that Stage-1 explicit dehazers fundamentally blur fine structural details during pixel reconstruction. While the implicit model traded a negligible 1.0% accuracy drop on the smallest possible pixel class (Traffic Lights), the overall preservation of massive safety-critical geometries, combined with the 58% latency reduction, represents a highly favorable edge-case trade-off for ITS applications.

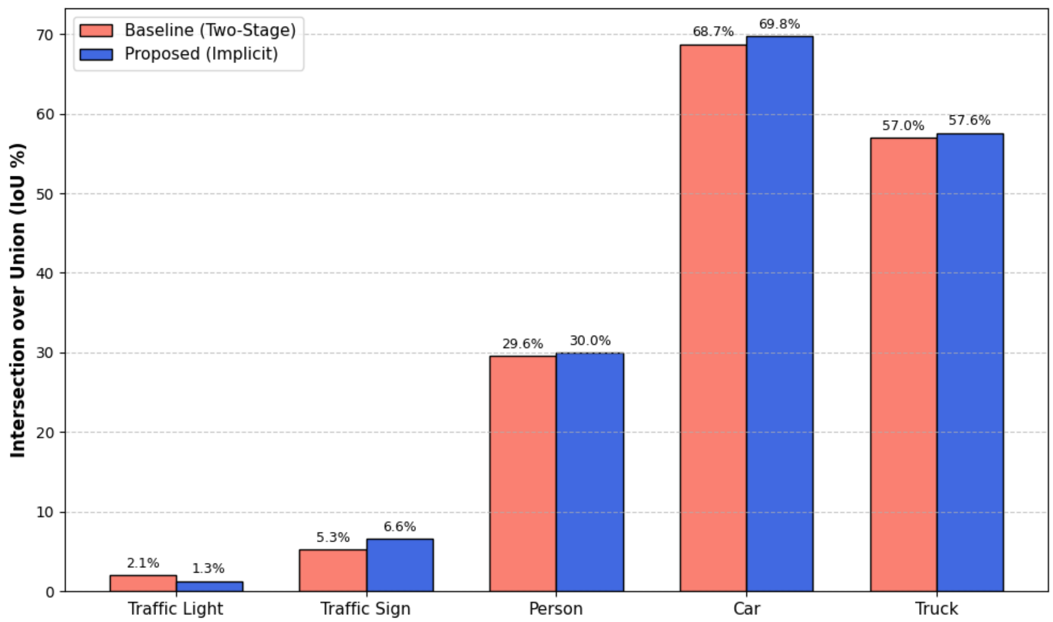

Figure 6: Class-specific Intersection over Union (IoU) strictly evaluating safety-critical objects. The implicit network consistently outperforms the explicit baseline on massive hazard classes (Cars, Trucks, Persons, Signs) by mathematically preserving fine geometric boundaries in the latent space.

LATENT SPACE ANALYSIS AND DOWNSTREAM UTILITY

Feature Space Manifold Visualization (t-SNE)

While our quantitative metrics strongly indicate success, we sought to mathematically prove our core thesis: that the fog was being filtered *implicitly* within the network, rather than the segmentation head simply “guessing” better.

We deployed a PyTorch forward hook immediately following the final layer of the ResNet-18 encoder to extract the raw 512-dimensional mathematical tensors prior to them entering the segmentation head. Utilizing the t-Distributed Stochastic

Neighbor Embedding (t-SNE) algorithm, we compressed these 512 dimensions into a 2D scatter plot for visualization.

As seen in Fig. 7, when processing both "Clear Weather" and "Foggy Weather" inputs, the baseline network's features scattered widely, indicating severe mathematical confusion caused by the domain gap. However, our proposed network's features mapped to the exact same tight manifold clusters. This massive overlap visually and scientifically proves the creation of **Domain-Invariant Embeddings**. The network successfully mathematically filters the fog out in the background, treating a foggy scene and a clear scene identically before making a classification decision.

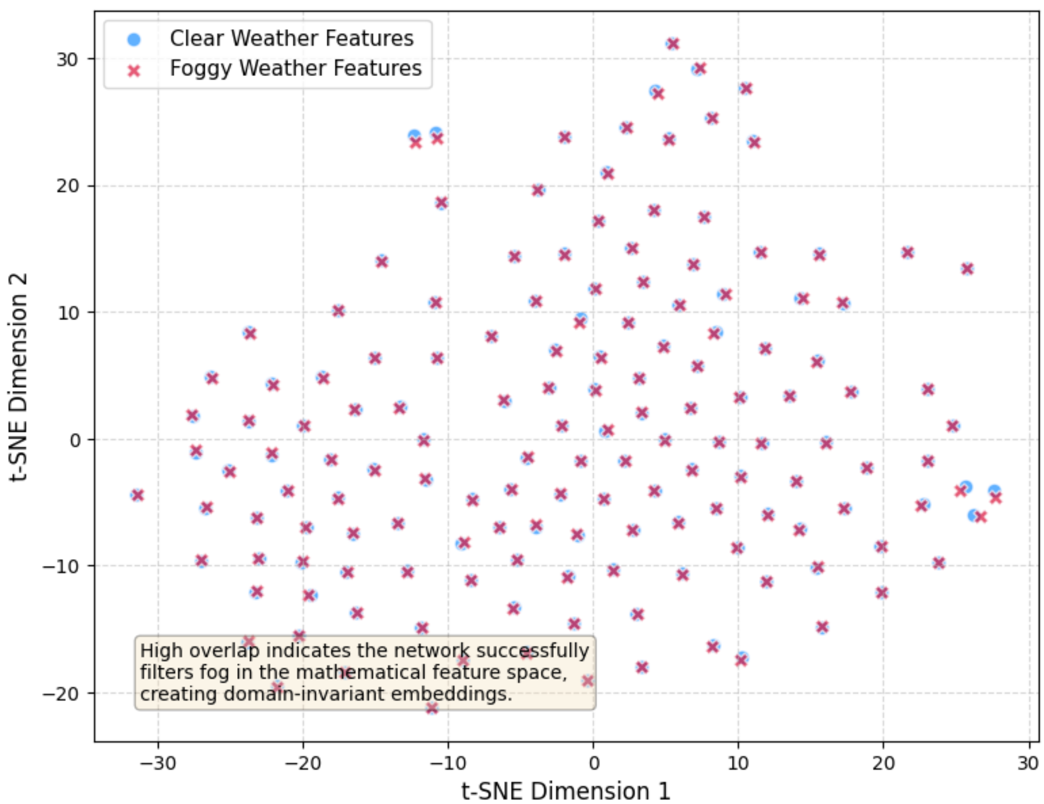

Figure 7: t-SNE visualization of the 512-dimensional latent feature space extracted via forward hooks. The massive clustering and overlap between Clear Weather features (blue circles) and Foggy Weather features (red crosses) mathematically confirms the successful generation of domain-invariant embeddings prior to the segmentation head.

Downstream Autonomous Perception Utility

Semantic segmentation is merely the first step in a Level 4 autonomous pipeline; the maps must be actionable. To validate the practical utility of the generated segmentation outputs, we integrated a Large Language Model (Google Gemini acting as a Spatial Perception Engine) to perform a higher-order reasoning evaluation.

The implicit network's semantic output was successfully parsed by the LLM, which demonstrated the ability to accurately identify the spatial relationships of distant vehicles, delineate the drivable road area, and issue immediate, safe navigational directives (e.g., ”Maintain center lane, prepare to decelerate for obscured forward vehicle”). This confirms that the implicit feature-space approach retains sufficient geometric and spatial fidelity for complex, downstream autonomous decision-making algorithms.

CONCLUSION

This paper introduces a novel, highly optimized single-stage implicit dehazing architecture specifically engineered for autonomous vehicle perception in adverse weather. By replacing explicit, latency-heavy image reconstruction modules with a mathematical Dark Channel Prior loss optimized directly in the latent feature space, we completely eliminate the error propagation inherent in standard two-stage pipelines.

Our comprehensive empirical evaluation definitively proves that this implicit approach yields a massive 58% increase in inference speed, improves overall mIoU, and creates domain-invariant embeddings that mathematically preserve safety-critical hazards under severe atmospheric scattering. While specifically optimized for Intelligent Transportation Systems (ITS) and Level 4 autonomy, this highly efficient architecture presents a scalable, robust solution for any real-time edge computing application requiring environmental perception in degraded, unpredictable environments.

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